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Stand, method of manufacturing stand, and keyboard-stand set

Abstract

A stand supports an electronic apparatus in an erect state. The stand includes: a supporting portion that has a supporting surface which supports a rear surface of the electronic apparatus; a leg portion that is rotatably connected to the supporting portion via a hinge; a flexible member that has flexibility, that is formed in a sheet shape, and that is provided along one edge of the supporting portion so as to protrude from the one edge; a surface material that has flexibility and forms front surfaces of the supporting portion and the flexible member; a first bonding layer that fixes the surface material onto the supporting portion; and a second bonding layer that is formed to have a lower coating density of a bonding adhesive than the first bonding layer and fixes the surface material onto the flexible member.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2011/0174533	12/2010	Nagano	29/832	H03H 9/1021
2013/0088818	12/2012	Yamaguchi	361/679.01	C09J 7/30
2013/0194516	12/2012	Hayashiyama	361/752	H04N 5/655
2014/0146451	12/2013	Aoki	361/679.01	H03H 9/1057
2015/0190114	12/2014	Ninomiya	600/437	A61B 8/467
2020/0347987	12/2019	Lee	N/A	B29C 48/022
2023/0292565	12/2022	Morino	N/A	H04M 1/0268

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2018105961	12/2017	JP	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a stand that supports an electronic apparatus in an erect state, a method of manufacturing the stand, and a keyboard-stand set.

Description of the Related Art

(2) In recent years, thin electronic apparatuses such as laptop PCs and tablet PCs, each of which has a touch panel type display and does not have a physical keyboard, have become widespread. Regarding the type of the electronic apparatuses, it is easy to carry the electronic apparatus since the input work may be performed through the touch panel. As a result, it is easy to operate the electronic apparatus. However, it is difficult for such an electronic apparatus to be used independently like a general clamshell type laptop PC. Therefore, the present applicant has proposed a kickstand-type stand that supports an electronic apparatus in an erect state in Japanese Unexamined Patent Application Publication No. 2018-105961.

SUMMARY OF THE INVENTION

- (3) In the stand as described above, for example, a member having flexibility may be provided at a part that supports a side surface of the electronic apparatus. Further, it is assumed that such a member having flexibility will be used as a flexible hinge for connecting the electronic apparatus or other device and the stand. Therefore, it is necessary for the member having flexibility to have high pliability in order to ensure operability at the time of bending or the like.
- (4) The present invention has been made in view of the above-described problems, and it is an object thereof to provide a stand capable of improving the pliability of a member having flexibility, a method of manufacturing the stand, and a keyboard-stand set.
- (5) According to a first aspect of the present invention, there is provided a stand that supports an electronic apparatus in an erect state. The stand includes: a supporting portion that has a supporting surface which supports a rear surface of the electronic apparatus; a leg portion that is rotatably connected to the supporting portion via a hinge; a flexible member that has flexibility, that is formed in a sheet shape, and that is provided along one edge of the supporting portion so as to protrude from the one edge; a surface material that has flexibility and forms front surfaces of the supporting portion and the flexible member; a first bonding layer that fixes the surface material onto the supporting portion; and a second bonding layer that is formed to have a lower coating density of a bonding adhesive than the first bonding layer and fixes the surface material onto the flexible member.
- (6) According to a second aspect of the present invention, there is provided a method of manufacturing a stand that supports an electronic apparatus in an erect state. The method includes: a bonding portion formation step of providing a bonding adhesive on an inner surface of a surface material formed in a sheet shape and forming a first bonding portion and a second bonding portion which has a lower coating density of the bonding adhesive than the first bonding portion; a supporting portion formation step of forming a supporting portion that supports a rear surface of the electronic apparatus by bonding the surface material to a plate-like member via the first bonding portion, after the bonding portion formation step; a flexible member formation step of forming a flexible member, which has flexibility, which is formed in a sheet shape, and which is provided along one edge of the supporting portion so as to protrude from the one edge, by bonding two surface materials via the second bonding portion, after the bonding portion formation step; and a leg portion mount step of rotatably mounting a leg portion on the supporting portion via a hinge, after the supporting portion formation step.
- (7) According to a third aspect of the present invention, there is provided a keyboard-stand set including: a keyboard that is for wirelessly inputting to an electronic apparatus; and a stand that is for supporting the electronic apparatus in an erect state. The stand includes a supporting portion that has a supporting surface which supports a rear surface of the electronic apparatus, a leg portion that is rotatably connected to the supporting portion via a hinge, a flexible member that has flexibility, that is formed in a sheet shape, that is provided along one edge of the supporting portion so as to protrude from the one edge, and that functions as a flexible hinge which rotatably connects the stand and one edge of the keyboard, a surface material that has flexibility and forms front surfaces of the supporting portion and the flexible member, a first bonding layer that fixes the surface material onto the supporting portion, and a second bonding layer that is formed to have a lower coating density of a bonding adhesive than the first bonding layer and fixes the surface material onto the flexible member.
- (8) According to the above-described aspects of the present invention, pliability of the member having flexibility can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a plan view of a keyboard-stand set according to an embodiment.
- (2) FIG. 2 is an exploded perspective view of a keyboard and a stand as viewed from a rear side.
- (3) FIG. 3 is a front view of the stand.
- (4) FIG. 4 is a perspective view of the stand in an erect state.
- (5) FIG. 5 is a schematic perspective view illustrating a state where an electronic apparatus is placed vertically and stands against the stand.
- (6) FIG. 6 is a schematic cross-sectional view taken along the line VI-VI in FIG. 3.
- (7) FIG. 7 is a schematic developed view of a surface material constituting a supporting portion and a flexible member.
- (8) FIG. 8 is a flowchart illustrating a procedure of a method of manufacturing the stand.
- (9) FIG. 9 is a front view of a stand according to a modification example.
- (10) FIG. 10 is a partially enlarged schematic perspective view illustrating a state where the electronic apparatus is placed vertically and stands against the stand illustrated in FIG. 9.

DETAILED DESCRIPTION OF THE INVENTION

(11) Hereinafter, a stand and a keyboard-stand set according to embodiments of the present invention will be described in detail with reference to suitable embodiments and the accompanying drawings.

(12) FIG. 1 is a plan view of a keyboard-stand set **10** according to the embodiment. FIG. 2 is an exploded perspective view of a keyboard **12** and a stand **14** as viewed from a rear side. As illustrated in FIGS. 1 and 2, the keyboard-stand set **10** according to the present embodiment includes the keyboard **12** and the stand **14**.

(13) First, the keyboard **12** will be described.

(14) The keyboard **12** is a device that is for wirelessly inputting to the electronic apparatus. As the electronic apparatus, a portable computer that does not have a physical keyboard is mostly considered. Examples thereof include thin electronic apparatuses such as a foldable PC, a tablet PC, and a smartphone. Wireless communication performed between the keyboard **12** and the electronic apparatus includes, for example, Bluetooth (registered trademark).

(15) The keyboard **12** has a flat box shape having a general size. A plurality of keys **12a**, a touch pad **12b**, and a pointing stick **12c**, which are aligned, are provided on the upper surface of the keyboard **12**. The keyboard **12** illustrated in FIG. 1 is a compact type that does not have a numeric keypad, but may be a full-size type that has a numeric keypad, an independently disposed cursor key, or the like. The keyboard **12** has a battery. In the keyboard **12**, one or both of the touch pad **12b** and the pointing stick **12c** may be removed.

(16) In the keyboard **12**, a recessed portion **16** is formed on a rear side edge portion **12d**. The recessed portion **16** extends along the edge portion **12d** and is open toward the lower surface side. The recessed portion **16** has a height and a length by which the bar **30** to be described later can be inserted and fit into the recessed portion **16**. A plurality of magnets **18** are provided on an upper surface **16a** of the recessed portion **16**. In the case of the present embodiment, two magnets **18** are provided in the vicinity of each of both ends of the upper surface **16a** and one magnet **18** is provided in the central portion. Consequently, a total of five magnets **18** is provided (refer to FIG. 2). The number, positions, and sizes of the magnets **18** are not limited thereto. The magnet **18** is covered with a thin cover which forms the upper surface **16a**.

(17) Next, the stand **14** will be described.

(18) FIG. 3 is a front view of the stand **14**. FIG. 4 is a perspective view of the stand **14** in an erect state. FIG. 5 is a schematic perspective view illustrating a state where an electronic apparatus **50** is placed vertically and stands against the stand **14**. The stand **14** supports, for example, the thin electronic apparatus **50** as illustrated in FIG. 5 in an erect state. As illustrated in FIGS. 1 to 4, the stand **14** includes a supporting portion **24**, a leg portion **26**, a flexible member **28**, and a bar **30**.

(19) The supporting portion **24** is a thin plate member having a rectangular frame shape. The supporting portion **24** is formed in a rectangular frame shape by forming a rectangular opening portion **32** on an inside thereof. One surface of the supporting portion **24** is a supporting surface **24a** which a rear surface **50a** of the electronic apparatus **50** is in contact with and which supports the rear surface **50a**. The outer peripheral contour of the supporting portion **24** is substantially the same as the keyboard **12** in plan view. A plurality of magnets **34** are embedded at dispersed positions in the supporting portion **24**. Each magnet **34** faces the supporting surface **24a**. A magnet is provided on the rear surface **50a** of the electronic apparatus **50**, and the magnet **34** is provided at a position at which the magnet **34** is able to attract the magnet of the electronic apparatus **50**.

(20) The leg portion **26** is a rectangular plate which functions as a kickstand. The leg portion **26** is connected to the supporting portion **24** via a torque hinge **36** inside the opening portion **32** and is relatively rotatable about the supporting portion **24**. In the case of the present embodiment, a pair of torque hinges **36** on the left and right sides are provided on one long side frame **24b** of the supporting portion **24**. The long side frame **24b** forms an edge of the supporting portion **24** on a side far from the flexible member **28**.

(21) As illustrated in FIG. **1**, for example, the torque hinge **36** is connected to the long side frame **24b** by using a bracket **36a** for the supporting portion **24**, and is connected to a base edge **26b** by using a bracket **36b** for the leg portion **26**. The brackets **36a** and **36b** are thin metal plates, for example, and are fixed to the supporting portion **24** and the leg portion **26** by using screws, double-sided tape, a bonding adhesive, or the like.

(22) The leg portion **26** is rotatable at least between a first angular position and a second angular position. The first angular position is a position at which the leg portion **26** is accommodated inside the opening portion **32** and the opening portion **32** is closed without an approximate gap (refer to FIGS. **1** and **6**). At the first angular position, the stand **14** is configured such that the supporting portion **24** and the leg portion **26** have a thin plate shape as a whole.

(23) The second angular position is a position at which a tip edge **26a** on a side of the leg portion **26** opposite to the torque hinge **36** side is in contact with a placement surface **G** and the stand **14** is erected (refer to FIG. **4**). The placement surface **G** is, for example, an upper surface of a desk or table. The second angular position is, for example, a position at which the surface normal direction of the supporting portion **24** and the surface normal direction of the leg portion **26** are approximately 90 degrees (refer to FIG. **3**). In addition, the leg portion **26** is connected to the base edge **26b** on a side opposite to the tip edge **26a** via the torque hinge **36**.

(24) The leg portion **26** is able to function as a kickstand even at an angle smaller than the angle at the second angular position, for example, an angle of about 30 degrees illustrated in FIG. **5**. The leg portion **26** is able to function as a kickstand at an angle larger than the angle at the second angular position, for example, approximately 120 degrees. The torque hinge **36** generates a predetermined rotary torque in a case where the torque hinge **36** is set at least an angle at which the leg portion **26** functions as a kickstand, for example, 30 degrees or more. Thereby, the leg portion **26** can be kept at any angle with respect to the supporting portion **24**. In addition, the stand **14** according to the present embodiment is configured such that the leg portion **26** opens in substantially only one direction with respect to the supporting portion **24**. However, the stand **14** maybe configured such that the leg portion **26** opens in both directions. The leg portion **26** may have a shape other than the rectangular shape.

(25) As illustrated in FIGS. **1** and **4**, the stand **14** has a plurality of first magnets **38N** and **38S** embedded in the supporting portion **24** and a plurality of second magnets **39N** and **39S** embedded in the leg portion **26**. The magnets **38N**, **38S**, **39N**, and **39S** serve as stoppers for attracting each other and keeping the leg portion **26** accommodated (first angular position). In the description of the present embodiment, the first magnet **38N** is disposed in a form in which a desired attraction surface of the magnet is the N pole, and the first magnet **38S** is disposed in a form in which a desired attraction surface of the magnet is the S pole. In addition, the same applies to second

magnets **39N** and **39S**. It is apparent that the number of the provided magnets **38N**, **38S**, **39N**, and **39S** and the order of arrangement of the magnets **38N**, **38S**, **39N**, and **39S** are not limited to those illustrated in FIGS. **1** and **4**.

(26) As illustrated in FIGS. **1** to **5**, the flexible member **28** is a sheet-like member having flexibility. The flexible member **28** is provided along an outer edge of a long side frame **24c** on a side opposite to the long side frame **24b**, and has a band shape protruding from the outer edge thereof. The flexible member **28** has a total length that is slightly shorter than a total length of the long side frame **24c**, and a width capable of supporting a side surface **50b** of the electronic apparatus **50**.

(27) The flexible member **28** is provided with a plurality of anti-slip members **51** on a first surface **28a** continuous from the supporting surface **24a**. The anti-slip member **51** extends along the longitudinal direction of the first surface **28a**, and is, for example, a rod-like rubber. The anti-slip member **51** of the present embodiment is not provided on a second surface **28b**, which is the rear surface of the first surface **28a**, but may also be provided on the second surface **28b**. In the in-use state posture of the stand **14** illustrated in FIGS. **4** and **5**, the flexible member **28** is bent from the long side frame **24c** and placed on the placement surface **G** in a state where the supporting portion **24** is kept at an appropriate angle.

(28) The bar **30** is provided along a tip edge **28d** opposite to a base edge **28c** on the supporting portion **24** side of the flexible member **28**. The bar **30** is a material that attracts the magnet, and is a cylindrical rod of an iron material (including an alloy such as a stainless steel material) in the present embodiment. The bar **30** is slightly longer than the flexible member **28**, and portions protruding from the flexible member **28** are covered with caps **30a**. The bar **30** is a part to be inserted into the recessed portion **16**, and has a thickness and a length by which the bar **30** can be inserted into the recessed portion **16**. The lengths of the recessed portion **16** and the bar **30** are substantially the same, and the bar **30** is not displaced in the elongated direction inside the recessed portion **16**. The tip edge **28d** of the flexible member **28** may be connected to the keyboard **12** in a non-detachable manner.

(29) FIG. **6** is a schematic cross-sectional view taken along the line VI-VI in FIG. **3**. As illustrated in FIG. **6**, in the stand **14** of the present embodiment, the front surfaces of the supporting portion **24** and the flexible member **28** are formed of a surface material **40**, and the front surface of the leg portion **26** is formed of a surface material **41**. The surface materials **40** and **41** are fabric materials such as suedes, felts, or leathers, and each has some elasticity in the thickness direction. The surface materials **40** and **41** of the present embodiment are made of suede.

(30) First, the supporting portion **24** is constituted of a plate-like member **42** which has a rectangular frame shape provided with the opening portion **32** on the inside thereof, and the surface material **40** which covers substantially the entire front surface of the plate-like member **42**. The plate-like member **42** may be formed of a hard resin plate. The plate-like member **42** may be formed of a metal plate, a hard rubber plate, or the like. The surface material **40** is fixed onto the front surface of the plate-like member **42** via a first bonding layer **44**, and covers the plate-like member **42** so as to wrap around the plate-like member **42**.

(31) The flexible member **28** is configured such that the two surface materials **40** and **40** are bonded via the second bonding layer **45**. Thereby, the flexible member **28** is formed as a sheet-like member having flexibility and pliability. As schematically illustrated in an enlarged view enclosed by a one-dot chain line in FIG. **6**, the second bonding layer **45** has an uneven shape **46a** or a hole shape **46b** formed at each location. On the other hand, the first bonding layer **44** does not form such an uneven shape **46a** or a hole shape **46b**. That is, the second bonding layer **45** has a lower coating density of the bonding adhesive than the first bonding layer **44**. However, the details will be described later. Further, the second bonding layer **45** of the present embodiment has a single-layer structure. That is, the second bonding layer **45** has a structure in which the two surface materials **40** and **40** are bonded in a state where the bonding adhesive is provided on only one inner surface **40a** of the two surface materials **40** and **40** to be bonded to each other and the bonding adhesive is not

provided on the other inner surface **40a**.

(32) The surface material **40** of the present embodiment is formed of a single sheet, and the front surfaces of the supporting portion **24** and the flexible member **28** are formed of the single sheet. Specifically, the surface material **40** extends from the supporting surface **24a** of the supporting portion **24** to the first surface **28a** of the flexible member **28**, is wound around the outer periphery of the bar **30**, and is folded back. The surface material **40** is bonded to the bar **30** via the third bonding layer **48**. The surface material **40** folded back at the bar **30** extends from the second surface **28b** of the flexible member **28** to a rear surface **24d** of the supporting portion **24**, which is opposite to the supporting surface **24a**. Further, the surface material **40** is bent from the supporting surface **24a** side to the rear surface **24d** side of the supporting portion **24** at the outer peripheral edge portion of the supporting portion **24**, and also covers the outer peripheral side surface of the supporting portion **24**. The flexible member **28** may have a configuration, in which the surface material **40** is bonded to the front surface of the flexible sheet such as a resin serving as a core material.

(33) As the surface material **40**, for example, two sheets may be used, and the supporting surface **24a** and the first surface **28a** and the rear surface **24d** and the second surface **28b** may be formed of different sheets from each other. Further, as the surface material **40**, the supporting portion **24** and the flexible member **28** may be formed of different sheets from each other.

(34) Next, the leg portion **26** includes a rectangular plate-like member **26c** and a surface material **41** that covers substantially the entire front surface of the plate-like member **26c**. The plate-like member **26c** can be formed of a plate which is made of the same or similar material as the plate-like member **42** described above. The surface material **41** is fixed onto the front surface of the plate-like member **26c** via a leg portion bonding layer **49**, and covers the plate-like member **26c** so as to wrap around the plate-like member **26c**. The surface material **41** is formed of a single sheet, and the single sheet forms the front surface of the leg portion **26**. For example, the surface material **41** may be configured to cover each surface of the plate-like member **26c** with two sheets.

(35) Next, one procedure of the method of manufacturing the stand **14** will be described.

(36) FIG. 7 is a schematic developed view of the surface material **40** constituting the supporting portion **24** and the flexible member **28**. FIG. 7 is a view of the surface material **40** as viewed from an inner surface **40a** side on which the bonding layers **44** and **45** are provided. FIG. 8 is a flowchart illustrating a procedure of a method of manufacturing the stand **14**.

(37) First, as illustrated in FIG. 7, the surface material **40** is placed such that the inner surface **40a** thereof faces in the front direction. The surface material **40** has a rear surface forming portion **40b** that forms the rear surface **24d** of the supporting portion **24**, a supporting surface forming portion **40c** that forms the supporting surface **24a**, a first surface forming portion **40d** that forms the first surface **28a** of the flexible member **28**, a second surface forming portion **40e** that forms the second surface **28b**, and a bar winding portion **40f** that is wound around the bar **30**. In addition, the supporting surface forming portion **40c** also covers the outer peripheral side surface of the plate-like member **42**. Therefore, the outer peripheral contour of the supporting surface forming portion **40c** is one size larger than that of the rear surface forming portion **40b**. Further, rectangular openings **40g**, which constitute the opening portion **32**, are respectively formed on the rear surface forming portion **40b** and the supporting surface forming portion **40c**.

(38) Then, the bonding adhesive is provided on the inner surface **40a** of the surface material **40** placed as illustrated in FIG. 7 to form a first bonding portion **44a** and a second bonding portion **45a** (step S1 in FIG. 8). In the present embodiment, a method is adopted in which a film-like bonding adhesive is laid so as to be pasted on the inner surface **40a** to thereby form each of the bonding portions **44a** and **44b**. The bonding portions **44a** and **44b** may be formed by a method of coating with a bonding adhesive other than the film-like bonding adhesive, and may be formed by, for example, a method of coating with a liquid bonding adhesive or the like.

(39) The first bonding portion **44a** is a part that forms the first bonding layer **44**. Therefore, the first

bonding portion **44a** is formed on each of the inner surface **40a** of the rear surface forming portion **40b** and the inner surface **40a** of the supporting surface forming portion **40c**.

(40) The second bonding portion **45a** is a part that forms the second bonding layer **45**. As described above, the second bonding layer **45** of the present embodiment has a single-layer structure.

Therefore, the second bonding portion **45a** is formed only on the inner surface **40a** of one second surface forming portion **40e** of the first surface forming portion **40d** and the second surface forming portion **40e** that are bonded to each other to form the flexible member **28**, and is not formed on the inner surface **40a** of the other first surface forming portion **40d**. As described above, the inner surface **40a** of the first surface forming portion **40d** is set as a non-bonding region R1 in which the bonding adhesive is not provided. On the other hand, the inner surface **40a** of the second surface forming portion **40e** is set as a bonding region R2 in which the bonding adhesive is provided, and the bonding region R2 forms the second bonding portion **45a**.

(41) As illustrated in FIG. 7, the second bonding portion **45a** is formed of a film-like bonding adhesive having a porous structure with a large number of hole portions **46**. The hole portion **46** is a part that forms the uneven shape **46a** or the hole shape **46b** described above in a case where the surface materials **40** are bonded to each other via the second bonding portion **45a** to form the second bonding layer **45**. The hole portion **46** is not provided in the first bonding portion **44a**. As described above, in the present embodiment, the second bonding portion **45a** has a large number of hole portions **46**. Therefore, the coating density of the bonding adhesive in the second bonding portion **45a** is smaller than that of the first bonding portion **44a**. As a result, the coating density of the second bonding layer **45** is smaller than that of the first bonding layer **44**.

(42) In the case of the present embodiment, the bonding adhesive, which forms the second bonding portion **45a**, has a film shape. Therefore, there is an advantage in that it is easy to form the hole portion **46** as compared with a case where the inner surface **40a** is coated with the liquid bonding adhesive. The shape of the hole portion **46** may be a shape other than a circular shape, and the second bonding portion **45a** may have a configuration, in which coating of the bonding adhesive having a mesh shape is performed, or the like.

(43) As illustrated in FIG. 7, the inner surface **40a** of the bar winding portion **40f** is provided with a film-like bonding adhesive without the hole portion **46** to form a third bonding portion **48a**, in a similar manner to the first bonding portion **44a** described above. The third bonding portion **48a** is a part that forms a third bonding layer **48**.

(44) Next, in step S2, the rear surface forming portion **40b** is bonded to the front surface of the plate-like member **42** on the rear surface **24d** side via the first bonding portion **44a**. Subsequently, in step S3, the bar winding portion **40f** is wound around the outer peripheral surface of the bar **30**, and the surface material **40** is folded back. Then, the two folded-back surface materials **40** and **40**, that is, the first surface forming portion **40d** and the second surface forming portion **40e** are bonded to each other via the second bonding portion **45a** to form the flexible member **28**. In addition, the anti-slip member **51** may be fixed onto the outer surface of the first surface forming portion **40d** in advance, or may be fixed onto the outer surface of the first surface forming portion **40d** after step S4 or S5.

(45) In step S4, the supporting surface forming portion **40c** is bonded to the front surface of the plate-like member **42** on the supporting surface **24a** side via the first bonding portion **44a**. Then, the outer peripheral edge portion of the supporting surface forming portion **40c** is appropriately bent to cover the outer peripheral side surface of the plate-like member **42**, thereby forming the supporting portion **24**. It should be noted that the order of performing in steps S2 and S4 may be reversed.

(46) Finally, the leg portion **26** is connected to the supporting portion **24** via the torque hinge **36** (step S5). Thereby, manufacturing of the stand **14** is completed. The leg portion **26** may be manufactured in advance by bonding the surface material **41** to the front surface of the plate-like member **26c** via the leg portion bonding layer **49** in a step separate from steps S1 to S4. At this

time, in a case where the surface material **41** constituting the leg portion **26** uses, for example, a material of the end in which the opening **40g** is cut out from the surface material **40**, the material cost can be reduced.

(47) Next, an example of how to use the keyboard-stand set **10** will be described.

(48) In the keyboard-stand set **10** including such a stand **14**, in a case where the stand **14** is mounted on the keyboard **12**, as illustrated in FIG. 2, in a case where the bar **30** is inserted into the recessed portion **16**, the bar **30** is attracted and fixed by the magnet **18**. Thereby, the keyboard **12** and the stand **14** are combined. In this state, the flexible member **28**, which connects the stand **14** and the keyboard **12**, is able to also function as a flexible hinge. That is, in the keyboard-stand set **10**, the stand **14** can be folded to be overlapped on the upper surface of the keyboard **12** by bending the flexible member **28**, and can be configured to be compact in a case of carrying the keyboard-stand set **10** or the like. On the other hand, in a case where the stand **14** is pulled from the keyboard **12**, the bar **30** comes out of the recessed portion **16** against the attraction force of the magnet **18**. Thereby, the stand **14** is removed from the keyboard **12**.

(49) As an example of the electronic apparatus **50** supported by the stand **14** according to the present embodiment, a so-called foldable PC can be exemplified as illustrated in FIG. 5. It is apparent that the stand **14** and the keyboard-stand set **10** are also applicable to other thin electronic apparatuses such as a tablet PC and a smartphone.

(50) The electronic apparatus **50** is a foldable PC, and includes a first chassis **54** and a second chassis **56** which are connected via a central hinge portion **52**. On the rear side of the rear surface **50a** of the electronic apparatus **50**, a foldable display **58** over substantially the entire surface from the first chassis **54** to the second chassis **56** is provided (refer to FIG. 10). The display **58** is, for example, a touch panel type organic EL. The electronic apparatus **50** can be input through a software keyboard displayed on the display **58**, and may be input from the keyboard **12**. In a case where the first chassis **54** and the second chassis **56** are folded at the hinge portion **52**, the display **58** is also folded.

(51) As illustrated in FIG. 5, the keyboard-stand set **10** is able to keep the electronic apparatus **50** in a state of being vertically erect. The first chassis **54** is provided with an attracting member such as a magnet or an iron material at a position corresponding to the magnet **34** of the supporting portion **24**. In such a case, the supporting portion **24** of the stand **14** supports only the first chassis **54**, the second chassis **56** protrudes from the stand **14**, and the position of the barycenter thereof is slightly high. However, the stand **14** attracts the first chassis **54** due to magnetic force, and thus the electronic apparatus **50** is stabilized. It is apparent that the keyboard-stand set **10** is able to also keep the electronic apparatus **50** in a state where the electronic apparatus **50** is erected horizontally. In such a case, although there is no attraction due to the magnetic force between the stand **14** and the first chassis **54**, the electronic apparatus **50** is stable since the position of the barycenter is low in a state where the chassis **54** and **56** are aligned on the left and right.

(52) The electronic apparatus **50** has the side surface **50b** which is in contact with the first surface **28a** of the flexible member **28** and is stabilized without slipping due to the action of the anti-slip member **51**. Since the leg portion **26** is able to be kept at an optional angle due to the action of the torque hinge **36**, the tilt angle of the supporting portion **24** can be adjusted for easy viewing of the display **58**.

(53) Since the keyboard **12** is a wireless type keyboard, a wired connection with the electronic apparatus **50** is unnecessary. Therefore, even in a case where the keyboard **12** is removed from the stand **14** and placed at a place some distance away, operations such as keyboard input are possible.

(54) As described above, the stand **14** of the present embodiment includes the supporting portion **24**, the leg portions **26**, the flexible member **28**, and the surface material **40** that forms the front surfaces of the supporting portion **24** and the flexible member **28**. Here, the bonding layers **44** and **45**, which fix the surface material **40**, are formed such that the coating density of the bonding adhesive in the second bonding layer **45** which fixes the surface material **40** onto the flexible

member **28** is smaller than that of the first bonding layer **44** which fixes the surface material **40** onto the supporting portion **24**.

(55) In such a manner, the stand **14** is configured such that the coating density of the bonding adhesive in the second bonding layer **45** in the flexible member **28**, which is required to be pliable, is smaller than the coating density of the bonding adhesive in the first bonding layer **44** in the supporting portion **24**. Therefore, the pliability of the flexible member **28** is prevented from being impaired by the second bonding layer **45**, which is cured after the surface material **40** is bonded. Therefore, high pliability can be obtained. As a result, for example, in a case where the flexible member **28** supports the side surface **50b** of the electronic apparatus **50** on the placement surface **G** or functions as a flexible hinge between the stand **14** and the keyboard **12**, high pliability is obtained. Therefore, the operability during bending and the followability to the placement surface **G** are improved. Further, the pliability of the flexible member **28** is improved. Therefore, it is possible to prevent the bar **30** from falling off from the magnet **18** in a case where the stand **14** is folded to be overlapped on the upper surface of the keyboard **12** by bending the flexible member **28**. Moreover, the pliability of the flexible member **28** is improved by using the second bonding layer **45** that is not visible in appearance. Therefore, it is not necessary to provide a structure for improving pliability of a notch on the outer surface or the like, for example. On the other hand, the coating density of the bonding adhesive in the first bonding layer **44** in the supporting portion **24**, which is required to be rigid, is higher than the coating density of the bonding adhesive in the second bonding layer **45**. Therefore, the stability of the stand **14** in a case where the stand **14** is erected is improved. Thus, it is possible to further suppress the wobbling of the electronic apparatus **50** in a case where the electronic apparatus **50** is erected.

(56) In particular, the flexible member **28** of the present embodiment is configured such that two surface materials **40** and **40**, which are formed of a fabric material such as suede, are bonded to each other via the second bonding layer **45**. For this reason, in the flexible member **28**, the coating density of the bonding adhesive in the second bonding layer **45** is low. Therefore, the pliability originally possessed by the material of the surface material **40** can be prevented from being impaired, and higher pliability can be ensured.

(57) At this time, the second bonding layer **45** has a single-layer structure in which the bonding adhesive is provided only on one of the two surface materials **40** and **40**. Therefore, the flexible member **28** is able to suppress the thickness of the second bonding layer **45** to the minimum, and the pliability is further improved. In addition, the above description shows the example of the manufacturing method in which no bonding adhesive is provided on the first surface forming portion **40d** of the surface material **40** that forms the first surface **28a** of the flexible member **28** and the bonding adhesive is provided on only the second surface forming portion **40e** that forms the second surface **28b**. However, the bonding adhesive may be provided on any of the forming portions **40d** and **40e**. However, the first surface forming portion **40d** is provided with the anti-slip member **51** on the outer surface thereof. Therefore, in a case where the bonding adhesive is provided on the first surface forming portion **40d**, the bonding pressure is uneven due to the unevenness of the anti-slip member **51** in bonding between the forming portions **40d** and **40e**. Thus, there is also a possibility that the forming portions **40d** and **40e** may not be evenly bonded. In this respect, in the present embodiment, the bonding adhesive is provided to the second surface forming portion **40e** which does not have the anti-slip member **51**. Therefore, there is an advantage that occurrence of such a defect can be suppressed.

(58) FIG. **9** is a front view of a stand **14A** according to a modification example. FIG. **10** is a partially enlarged schematic perspective view illustrating a state where the electronic apparatus **50** is placed vertically and stands against the stand **14A** illustrated in FIG. **9**. In addition, in FIGS. **9** and **10**, the same reference numerals as the reference numerals illustrated in FIGS. **1** to **8** indicate the same or similar configurations. Therefore, the same or similar functions and effects are achieved, and detailed description thereof will be omitted.

(59) In the stand **14A** illustrated in FIGS. **9** and **10**, the flexible member **28** is different from the stand **14** described above in that the flexible member **28** includes the cutout hole **60**. The cutout hole **60** is, for example, a rectangular cutout hole which is formed at a substantially center in the longitudinal direction of the flexible member **28**, and has a certain length and width.

(60) As illustrated in FIG. **10**, for example, a pair of microphone holes **62a** and **62b** are formed on the side surface **50b** of the electronic apparatus **50**. The microphone holes **62a** and **62b** are fine hole portions for passing external sound toward the microphone provided in the second chassis **56** of the electronic apparatus **50**. The cutout hole **60** is provided at a position at which the side surface **50b** overlaps with the microphone holes **62a** and **62b** in a case where the side surface **50b** is placed on the first surface **28a** of the flexible member **28**, and does not cover the microphone holes **62a** and **62b**. The reference numeral **64** in FIG. **10** is a magnet which is provided on an upper surface of the electronic apparatus **50** and which attracts the chassis **54** and **56** in order to keep the folded state in a case of folding between the chassis **54** and **56**.

(61) Therefore, in the stand **14A**, in a case where the side surface **50b** is placed on the first surface **28a** of the flexible member **28** to support the electronic apparatus **50**, for example, the bar **30** may be attracted to the magnet **64**, and may cover the side surface **50b** such that the flexible member **28** is wound. Even in such a case, the microphone holes **62a** and **62b** are exposed through the cutout hole **60**.

(62) That is, in the stand **14** which does not have the cutout hole **60**, in a case where the flexible member **28** causes fine vibration due to an angle adjustment of the leg portion **26**, a touch operation with respect to the display **58**, or the like, there is also a concern that the vibration has an effect on the microphone holes **62a** and **62b**. In such a case, the gain of the microphone of the electronic apparatus **50** may decrease, and there is a possibility that ambient voice or the like is unlikely to be appropriately collected. In this respect, in the stand **14A**, the microphone holes **62a** and **62b** are exposed through the cutout hole **60**, and thus such a decrease in gain can be suppressed. In particular, the stand **14A** is provided with the flexible member **28** having high pliability, like the stand **14**. Therefore, the flexible member **28** easily wind around the side surface **50b**, and the cutout hole **60** is further effective. The shape and arrangement of the cutout hole **60** are appropriately changeable as long as the shape or arrangement does not cover the microphone hole of the electronic apparatus on which the stand **14A** is supposed to be used.

(63) The present invention is not limited to the above-described embodiment, and there is no doubt that the present invention can be freely changed without departing from the gist of the present invention.

(64) In the configuration exemplified in the above description, the stands **14** and **14A** are connected to the keyboard **12**. However, the stands **14** and **14A** may be used as a single stand that is not connected to the keyboard **12**. In such a case, in the stands **14** and **14A**, the bar **30** may be used to be attracted to the magnet **64** of the electronic apparatus **50**, similarly to, for example, the stand **14A** illustrated in FIG. **10**.

Claims

1. A stand that is configured to support an electronic apparatus in an erect state, the stand comprising: a supporting portion having a supporting surface that is configured to support a rear surface of the electronic apparatus; a leg portion rotatably connected to the supporting portion via a hinge; a flexible member in a sheet shape, and that is along and protrudes from one edge of the supporting portion; a surface material that is flexible and forms front surfaces of the supporting portion and of the flexible member; a first bonding layer that fixes the surface material onto the supporting portion; and a second bonding layer that has a lower coating density of a bonding adhesive than that of the first bonding layer and fixes the surface material onto the flexible member.

2. The stand according to claim 1, wherein the second bonding layer includes an uneven shape or a hole shape.
3. The stand according to claim 1, wherein the flexible member includes two surface materials that are bonded together with the second bonding layer, and the second bonding layer has a single-layer structure.
4. The stand according to claim 3, further comprising: a bar along a tip edge of the flexible member and on a side opposite to the one edge side of the supporting portion and that is of a material capable of being attracted to a magnet, wherein the flexible member includes the two surface materials, which are wound around the bar and folded back, with the second bonding layer.
5. The stand according to claim 1, wherein the flexible member has a width configured to support a side surface of the electronic apparatus, wherein the electronic apparatus has a microphone hole on the side surface, and the flexible member has a cutout hole that exposes the microphone hole without covering the microphone hole, when the flexible member supports at least the side surface of the electronic apparatus.
6. A method of manufacturing a stand that is configured to support an electronic apparatus in an erect state, the method comprising: a bonding portion formation step of providing a bonding adhesive on an inner surface of a surface material formed in a sheet shape and forming a first bonding portion and a second bonding portion, wherein the second bonding portion has a lower coating density of a bonding adhesive than that of the first bonding portion; a supporting portion formation step of forming a supporting portion that is configured to support a rear surface of the electronic apparatus by bonding the surface material to a plate-like member via the first bonding portion, after the bonding portion formation step; a flexible member formation step of forming a flexible member in a sheet shape, and which is provided along one edge of the supporting portion so as to protrude from the one edge, by bonding two surface materials via the second bonding portion, after the bonding portion formation step; and a leg portion mount step of rotatably mounting a leg portion on the supporting portion via a hinge, after the supporting portion formation step.
7. The method of manufacturing the stand according to claim 6, wherein in the bonding portion formation step, the first bonding portion is formed by providing the bonding adhesive in a film shape on an inner surface of the surface material, and the second bonding portion is formed by the bonding adhesive having holes and having the film shape on the inner surface of the surface material.
8. The method of manufacturing the stand according to claim 6, wherein in the bonding portion formation step, the second bonding portion is provided only on an inner surface of one surface material of the two surface materials which are bonded in the flexible member formation step, and an inner surface of the other surface material is not coated with the bonding adhesive, and in the flexible member formation step, the one surface material and the other surface material are bonded via the second bonding portion on the inner surface of the one surface material.
9. A keyboard-stand set comprising: a keyboard for wirelessly inputting to an electronic apparatus; and a stand for supporting the electronic apparatus in an erect state, wherein the stand includes: a supporting portion having a supporting surface that is configured to support a rear surface of the electronic apparatus, a leg portion rotatably connected to the supporting portion via a hinge, a flexible member in a sheet shape, that is along and protrudes from one edge of the supporting portion, and that functions as a flexible hinge which rotatably connects the stand and one edge of the keyboard, a surface material that is flexible and forms front surfaces of the supporting portion and of the flexible member, a first bonding layer that fixes the surface material onto the supporting portion, and a second bonding layer that has a lower coating density of a bonding adhesive than that of the first bonding layer and fixes the surface material onto the flexible member.
10. The keyboard-stand set according to claim 9, wherein a magnet is on one edge of the keyboard, the stand includes a bar that is along a tip edge of the flexible member on a side opposite to one

edge side of the supporting portion and that is of a material capable of being attracted to a magnet, and is configured to be attachable to and detachable from the keyboard, and the flexible member has two surface materials which are wound around the bar and folded back, and are bonded with the second bonding layer.
